Linux File Permissions & chmod

In Linux, file permissions control who can read, write, or execute a file or directory.

Each file/directory has three types of permissions and three categories of users.

TYPES OF PERMISSIONS

- Read (r):

  Files -> allows reading the contents.

  Directories -> allows listing its contents.

- Write (w):

  Files -> allows modifying or deleting contents.

  Directories -> allows creating, deleting, or renaming files inside.

- Execute (x):

  Files -> allows running it as a program/script.

  Directories -> allows entering (cd) into it.

PERMISSION GROUPS

- Owner (u): user who owns the file.
- Group (g): users in the file's group.
- Others (o): everyone else.

Example of ls -l output:

-rwxr-xr-- 1 prashant users 1024 Sep 25 19:00 script.sh

Breakdown:

- = file type

rwx = Owner: read, write, execute

r-x = Group: read, execute

r-- = Others: read only

NUMERIC REPRESENTATION

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rwx = 7 (4+2+1)

rw- = 6 (4+2)

r-x = 5 (4+1)

r-- = 4

-wx = 3

-w- = 2

--x = 1

--- = 0

Examples:

chmod 777 file -> everyone full access (unsafe)

chmod 644 file -> owner read/write, others read only

chmod 755 dir -> owner full, others list/enter

COMMON chmod USE CASES

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1. Give execute permission:

chmod +x hello.sh

./hello.sh -> runs script

2. Restrict access (private file):

chmod 600 secrets.txt -> only owner can read/write

3. Share executable:

chmod 755 app.sh -> owner rwx, others r-x

4. Allow group editing:

chmod 664 notes.txt -> owner+group rw-, others r--

USEFUL COMMANDS

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chmod +x file.sh -> Add execute for owner

chmod -x file.sh -> Remove execute

chmod u+x file.sh -> Owner execute only

chmod g+w file.txt -> Add write to group

chmod o-r file.txt -> Remove read from others

chmod u=rwx,g=rx,o=r f -> Explicit set perms

chmod ugo-rwx file.txt -> Remove all perms

chmod ugo+rwx file.txt -> Give all perms

chmod -R 755 /dir -> Recursive set perms

chmod --reference a b -> Copy perms from a to b

DEFAULT PERMISSIONS & umask

Default: files = 666 (rw-rw-rw-), dirs = 777 (rwxrwxrwx)

Formula: Default - umask = actual perms

Check umask:

$ umask

0022 -> removes write for group & others

Example:

File: 666 - 022 = 644 -> rw-r--r--

Dir: 777 - 022 = 755 -> rwxr-xr-x

Set umask:

umask 0077 -> New files = 600, dirs = 700 (owner only)

SUMMARY

- ls -l shows permissions
- chmod changes permissions
- umask defines default permissions
- Use symbolic (u+x) or numeric (755)
- Avoid 777 for sensitive files